

Kwabena Kingsley Kumah, Ph.D.

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Research Profile

Satellite precipitation scientist developing and evaluating retrievals and climate records where precipitation is difficult to observe: high latitudes, oceans, Antarctica, and data-sparse tropical regions. At the University of Arizona, contributes to NASA-supported GPCP research spanning AVHRR-based high-latitude retrieval, multi-reference ocean evaluation, and observation-constrained Antarctic snowfall assessment. Doctoral work developed commercial-microwave-link and Meteosat methods for rainfall monitoring in Sub-Saharan Africa.

Honors and Recognition

2026 AGU Open Science Recognition Prize – Global Precipitation Climatology Project Team. Member of the GPCP team recognized by the American Geophysical Union for long-term stewardship of open global precipitation data and trusted climate records. [Official AGU recipient page](#)

Professional Appointments

Postdoctoral Research Associate

2023–Present

Department of Hydrology & Atmospheric Sciences, University of Arizona, Tucson, AZ

- Develops and evaluates satellite precipitation retrievals and climate-data records, including AVHRR high-latitude retrieval and diagnostic evaluation for the published GPCP Version 3.3 record.
- Develops reusable algorithms, workflows, and toolsets for evaluating precipitation products, including GMI-GPROF, GPM IMERG, GPCP, ERA5, and MERRA-2 across global and regional scales, latitude and longitude bands, and surface states.
- Applies complementary reference datasets and a comprehensive evaluation framework—including distributional diagnostics (histograms and PDFs), categorical and continuous metrics, spatial diagnostics, and environmental stratification—to identify product strengths, biases, and retrieval sensitivities.

Visiting Scientist – EUMETSAT / H-SAF

2022–2023

European Organisation for the Exploitation of Meteorological Satellites, Darmstadt, Germany

Evaluated operational satellite precipitation products over Sub-Saharan Africa, with emphasis on diagnostic validation and uncertainty characterization.

Doctoral Researcher

2016–2022

Department of Water Resources, University of Twente, Enschede, Netherlands

Developed and published rainfall-detection and mapping methods combining commercial microwave-link attenuation with Meteosat Second Generation observations for data-sparse regions.

Education

Ph.D., GIS & Earth Observation – Hydrometeorology / Water Resources

2022

University of Twente, Faculty ITC, Enschede, Netherlands

Dissertation: *High-Spatiotemporal Resolution Rainfall Estimation from Satellite and Commercial Microwave Link Data.*

M.Sc., GIS & Earth Observation – Water Resources & Environmental Management

2016

University of Twente, Faculty ITC, Enschede, Netherlands

B.Sc., Environmental Science

2012

University of Cape Coast, Ghana

Selected Research Contributions

Satellite retrieval and climate records. Develops the University of Arizona High-latitude Infrared-based Precipitation Analysis (UA-HIPA), an AVHRR-based retrieval for regions poleward of 55°, including a physically informed correction for scan-angle-related infrared brightness-temperature bias. Contributes diagnostic evaluation supporting long-term precipitation records such as GPCP and IMERG.

Ocean and Antarctic precipitation evaluation. First author and lead analyst for two submitted studies: a multi-reference evaluation of GPCP, IMERG, ERA5, and MERRA-2 over oceans; and an observation-constrained Antarctic snowfall assessment combining GRACE/GRACE-FO storage change, ice discharge, basal melt, and sublimation with satellite and reanalysis products.

Snow and ice climate-data-record extension. Developed machine-learning and climatology-based methods using ERA5 surface variables to extend the GMA SI snow and ice record to 1980–1987. The peer-reviewed study and open dataset provide a retrospective record before the operational satellite era.

Operational rainfall monitoring from communication networks. Developed and operationalized a commercial-microwave-link rainfall retrieval and mapping workflow for Ghana. Gridded rainfall products and NetCDF delivery are implemented through a TAHMO-hosted API; current development adds link-path support and confidence layers for downstream evaluation and integration by Rainboo within TEMBO Africa activities.

Selected Peer-Reviewed Publications and Open Dataset

Behrangi, A., Huffman, G. J., Adler, R. F., Song, Y., **Kumah, K. K.**, Zandi, O., Bolvin, D. T., Nelkin, E. J., & Gu, G. (2026). GPCP Version 3.3: A refined global view of precipitation from satellite and gauge observations. *Journal of Hydrometeorology*. Published online August 26, 2026. doi:10.1175/JHM-D-26-0048.1

Kumah, K. K., Zandi, O., & Behrangi, A. (2025). Retrospective mapping of global snow and ice cover beyond the satellite observational era. *Earth and Space Science*. doi:10.1029/2024EA004171

Kumah, K. K., Maathuis, B. H. P., Hoedjes, J. C. B., & Su, Z. (2022). Near real-time estimation of high spatiotemporal resolution rainfall from cloud-top properties of MSG and CML rainfall intensities. *Atmospheric Research*, 279. doi:10.1016/j.atmosres.2022.106357

Kumah, K. K., Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2021). The MSG technique: Improving commercial microwave-link rainfall intensity using rain-area detection from Meteosat Second Generation. *Remote Sensing*, 13(16), 3274. doi:10.3390/rs13163274

Kumah, K. K., Maathuis, B. H. P., Hoedjes, J. C. B., Rwasoka, D. T., Retsios, B. V., & Su, B. Z. (2021). Rain-area detection in south-western Kenya using multispectral satellite data from Meteosat Second Generation. *Sensors*, 21(10), 3547. doi:10.3390/s21103547

David, N., Liu, Y., **Kumah, K. K.**, Hoedjes, J. C. B., Su, B. Z., & Gao, H. O. (2021). On the power of microwave communication data to monitor rain for agricultural needs in Africa. *Water*, 13(5), 730. doi:10.3390/w13050730

Kumah, K. K., Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2020). Combining MWL and MSG SEVIRI satellite signals for rainfall detection and estimation. *Atmosphere*, 11(9), 884. doi:10.3390/atmos11090884

Open dataset. **Kumah, K. K.**, Zandi, O., & Behrangi, A. (2024). Global snow and ice cover (1980–1987): Extended GMA SI dataset. University of Arizona Research Data Repository. doi:10.25422/azu.data.28012532.v1

Manuscripts Under Review and in Internal Review

SUBMITTED Kumah, K. K., Behrangi, A., Zandi, O., Gardner, A. S., Wiese, D. N., & Greene, C. A. (2026). Quantifying Antarctic snowfall accumulation using the latest ice discharge and spaceborne gravity observations: Comparison with reanalysis and satellite precipitation products.

SUBMITTED Kumah, K. K., Behrangi, A., Huffman, G. J., Adler, R. F., Gu, G., Song, Y., Zandi, O., Bolvin, D. T., Nelkin, E. J., Funk, C. C., & Peterson, P. (2026). A multi-reference assessment of ocean precipitation products by integrating PAL, buoys, atolls, and OceanRAIN.

IN INTERNAL REVIEW Kumah, K. K., Zandi, O., & Behrangi, A. (2026). Evaluating changes from GPROF Version 7 to Version 8 GMI precipitation retrievals over the contiguous United States.

Selected Conference Contributions

EGU General Assembly, Vienna, Austria 2026
Coauthor

Behrangi, A., et al., including **K. K. Kumah**. The newly released GPCP Version 3.3 daily and monthly products and future plans. EGU26-5885.

AGU Fall Meeting, New Orleans, LA 2025
Coauthor

Zandi, O., Pfreundschuh, S., **K. K. Kumah**, & Behrangi, A. University of Arizona High-latitude Infrared-based Precipitation Analysis Version 2 (UA-HIPA).

AGU Fall Meeting, Washington, D.C. 2024
Presenter and first author

Kumah, K. K., Zandi, O., & Behrangi, A. On the retrospective mapping of global snow and ice cover beyond the satellite observational era.

EUMETSAT Meteorological Satellite Conference, Brussels, Belgium 2022
Oral presenter and first author

Kumah, K. K. High spatial and temporal resolution rainfall estimation from MSG cloud-top properties using a random-forest algorithm trained with microwave-link-derived rainfall. Paper 107, Session S07-B.

KIT/HyMet CML Symposium, Garmisch-Partenkirchen, Germany 2019
Presenter and first author

Kumah, K. K. Commercial microwave links and Meteosat Second Generation observations for rainfall studies in Kenya. Short presentation and poster.

Professional Service

Peer-review contributions. Completed manuscript reviews for *Journal of Geophysical Research: Atmospheres*, *Earth System Science Data*, and *Remote Sensing*, including an invited review for *JGR: Atmospheres* in 2025.

Proposal review. Completed an external proposal review for the Swiss National Science Foundation.

Research community service. Working Group 3 member, COST Action CA20136 SmartSense.

Selected Media and Research Features

Mobile Phone Signals Hold Real-Time Rainfall Information – APS Physics Magazine, 2021.

ITC algorithm detects rainfall in rural Africa – University of Twente / ITC, 2022.

A First for Ghana: Turning Mobile Phone Signals into Near-Real Time Rainfall Maps – TEMBO Africa.

Technical Methods

Scientific computing and data systems: Python, NumPy, Pandas, Xarray, Dask, scikit-learn, Git, Linux/HPC workflows; NetCDF, HDF5, GRIB, and GeoTIFF.

Remote sensing and geospatial analysis: AVHRR, MSG SEVIRI, GPM/IMERG, GPCP, GRACE/GRACE-FO, ERA5, MERRA-2, radar-gauge products, GIS and raster processing.

Retrieval and evaluation: machine learning, satellite precipitation retrieval, multi-reference validation, uncertainty diagnostics, climate-data records, bias assessment, and reproducible production pipelines.